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Security Operations Center

Heart of Cyber Security

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Overview

- A Security Operations Center (SOC) is a centralized function within an organization dedicated to continuously monitoring, detecting, analyzing, and responding to cybersecurity threats.
- SOCs combine advanced technology, clearly defined processes, and skilled personnel to safeguard the organization's digital assets.

Core Functions

Continuous Monitoring

24/7 surveillance of networks, endpoints, applications, databases, and cloud environments for suspicious activity.

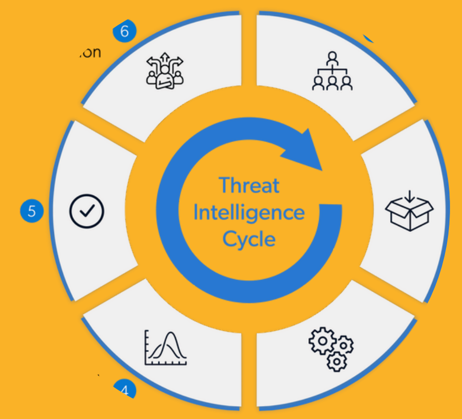
Threat Detection & Analysis

Collection, correlation, and real-time analysis of security events to spot anomalies and threats.

Incident Response

Investigation, triage, containment, and remediation of security incidents using tested protocols and workflows.

Core Functions



Threat Intelligence Integration

Use of up-to-date cyber threat intelligence to anticipate, recognize, and counter emerging tactics.

Proactive Threat Hunting

Analysts proactively search for hidden threats using hypothesis-driven methodologies.

Audit Logging & Compliance

Maintaining logs and evidential records for compliance obligations (e.g., GDPR, PCI DSS, NIST).

Methodology

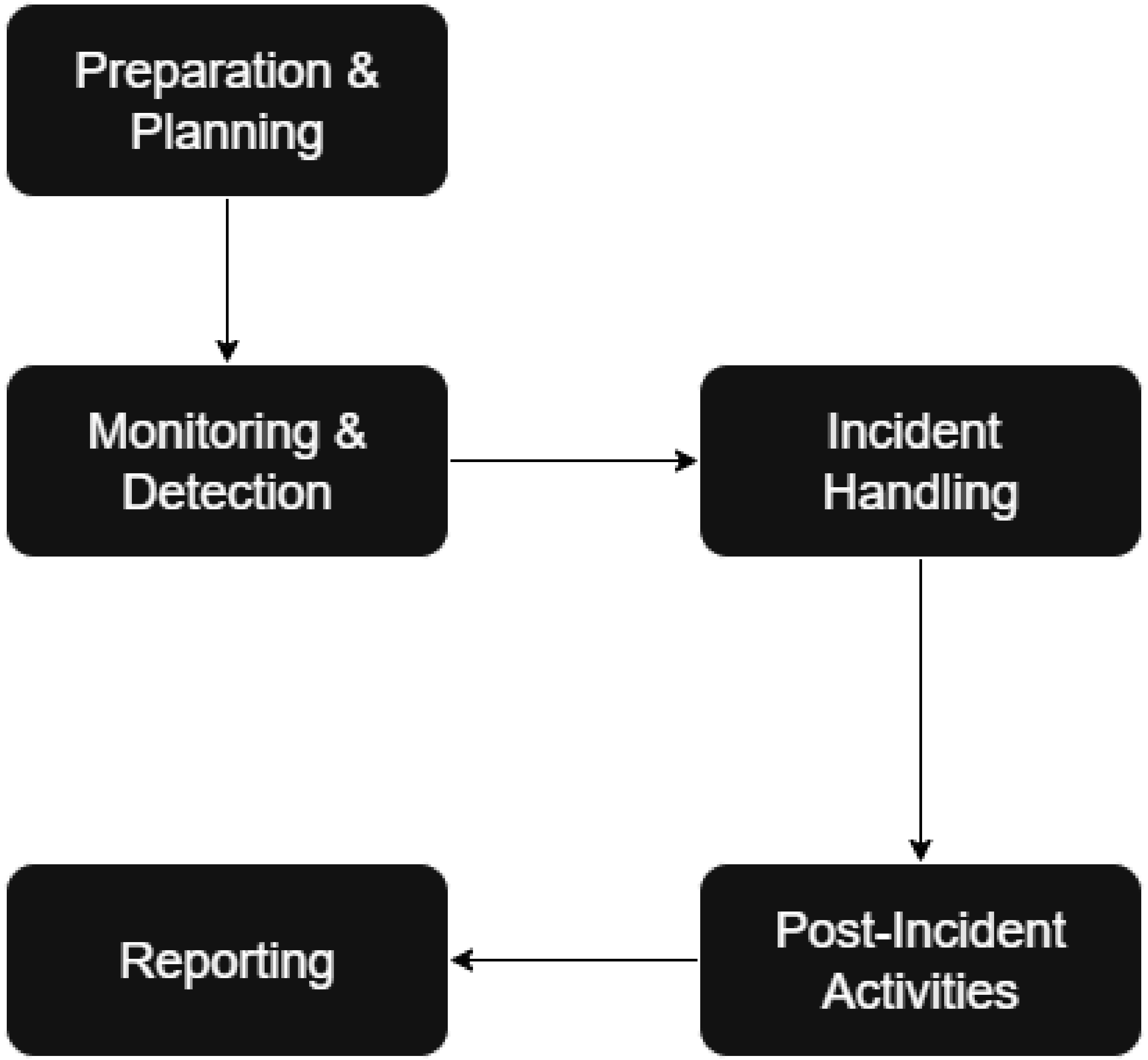
- **Preparation & Planning**
  - Understand the organizational environment.
  - Define roles, responsibilities, incident response plans.

- **Monitoring & Detection**
  - Deploy monitoring tools (SIEM, EDR, network sensors).
  - Tune alerting to balance detection.

- **Incident Handling**
  - Triage, analyze, and prioritize security incidents.
  - Contain, recover, and document each incident.

- **Post-Incident Activities**
  - Root cause analysis and lessons learned.
  - Continuous improvement of processes and defenses.

- **Reporting**
  - Regular dashboards and incident reports for stakeholders.



Yash Models

24/7

Provides continuous, round-the-clock security monitoring and incident response without any downtime.

16/7

Offers extended monitoring coverage for 16 hours a day, 7 days a week, balancing coverage and cost.

8/5

Operates during regular business hours (8 hours a day, 5 days a week), suitable for low-risk or smaller organizations.

Custom

Tailors monitoring hours and services to fit the unique needs and risk profiles of the organization.

Types of SOC

- **In-House SOC**
  - A fully owned and operated security operations center, managed by an organization’s own personnel.
  - Dedicated Physical or Virtual Facility
  - Direct Control
  - Tailored to Business needs

- **Outsourced SOC**
  - Security operations are fully or largely handled by an external Managed Security Service Provider (MSSP).
  - Third-Party Expertise and Infrastructure
  - Scalable and Rapid Deployment:

- **Hybrid SOC**
  - Combines resources from both the organization and an external MSSP for a flexible and collaborative approach.
  - Shared Responsibilities
  - Best of Both Worlds

SOC Team Composition

L1

- Monitoring Security Alerts
- Initial Incident Triage
- Basic Threat Analysis
- Escalation Procedure
- Documentation
- User Support

L2

- Advanced Threat Investigation
- Incident Response Coordination
- Incident Reporting and Post-Incident Analysis
- Mentoring L1 Analysts

L3

- Threat Hunting
- Expert Threat Analysis and Research
- Creating Detection Rules
- Providing Strategic Guidance
- Leading Incident Response
- Training and Development

L1 Team

Frontline defense, responsible for the bulk of day-to-day security monitoring and basic incident handling.

Key Responsibilities:

- Monitoring Security Alerts: Continuously track security alerts from SIEM tools and monitoring dashboards.
- Initial Incident Triage: Quickly assess the severity and relevance of incoming alerts, distinguishing false positives from real threats.
- Basic Threat Analysis: Perform preliminary investigation, gather information on suspicious activities, and categorize incidents.
- Escalation Procedure: Follow standard operating procedures (SOPs) to escalate incidents to higher levels if they require advanced expertise.
- Documentation: Record all actions, observations, and escalation outcomes for traceability and compliance.
- User Support: Assist organizational users with security-related questions, simple troubleshooting, and guidance on reporting incidents.

L2 Team

Handle complex incidents, deepen investigations, and ensure coordination for incident response activities.

Key Responsibilities:

- Advanced Threat Investigation: Perform in-depth analysis on escalated incidents—correlate logs, analyze malware, and identify attack paths.
- Incident Response Coordination: Lead response efforts for confirmed incidents, working with IT and business stakeholders for containment and recovery.
- Incident Reporting & Post-Incident Analysis: Create comprehensive reports detailing incident impacts, root causes, and recommended remediations.
- Mentoring L1 Analysts: Guide L1 staff, share expertise, and ensure adherence to best practices and processes.

L3 Team

High-level experts responsible for advanced threat activities, strategic improvements, and team development.

Key Responsibilities:

- Threat Hunting: Proactively search for hidden threats using hypothesis-driven techniques, custom queries, and threat intelligence.
- Expert Threat Analysis & Research: Conduct deep-dive analysis on new threat vectors, zero-day exploits, and advanced persistent threats.
- Creating Detection Rules: Develop and refine rules, signatures, and machine learning models to improve detection accuracy for SIEM and other monitoring systems.
- Leading Incident Response: Take charge of high-profile or complex security incidents—including those with legal or regulatory implications.
- Training & Development: Design training programs for team development, support certifications, and foster knowledge sharing.

Components & Tools of SOC

Security Info. & Event Management (SIEM)

- Aggregate, Correlate & Analyze Security Logs
- Splunk, IBM QRadar,

Security Orchestration, Automation & Response

- AI workflow builders, case summarization
- Cortex XSOAR, Torq

User & Entity Behavior Analytics (UEBA)

- Insider threat detection via behavioral baselining
- Exabeam, Varonis

Components & Tools of SOC

Threat Intelligence Platforms (TIP)

- Provide real-time threat context and indicators
- Anomali, ThreatConnect

Endpoint Detection & Response (EDR)

- Monitor and respond to endpoint threats
- CrowdStrike Falcon, SentinelOne

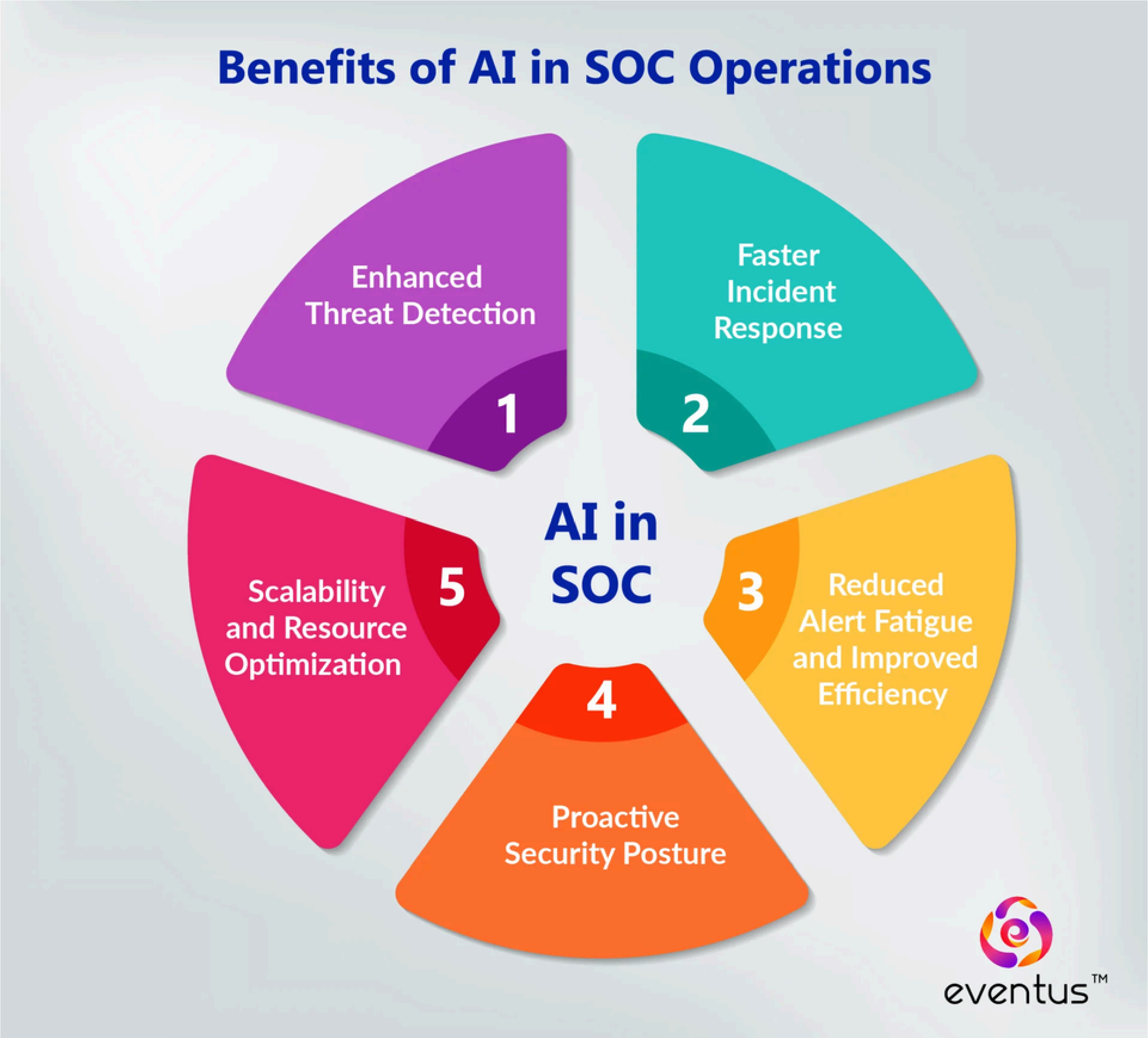
Sentinel in SOC

- Microsoft Sentinel is a leading **cloud-native** SIEM (Security Information and Event Management) and SOAR (Security Orchestration, Automation, and Response) platform.
- It provides **advanced threat detection**, **streamlined investigations**, and rapid, automated incident response—all centralized in one modern solution.

Feature	**Description**
Centralized Security Data	Single-pane view across cloud, hybrid, and on-prem systems
AI Threat Detection	Built-in machine learning and user/entity analytics for faster, more accurate alerts
Automation & Playbooks	SOAR capabilities automate routine and complex incident response
Cloud-Native Scalability	Rapid scaling, reduced capital expenditure, and flexible growth path
Investigation Workflows	Guided incident handling from triage to remediation
Deep Reporting	Customized dashboards, compliance tracking, and executive-ready summaries

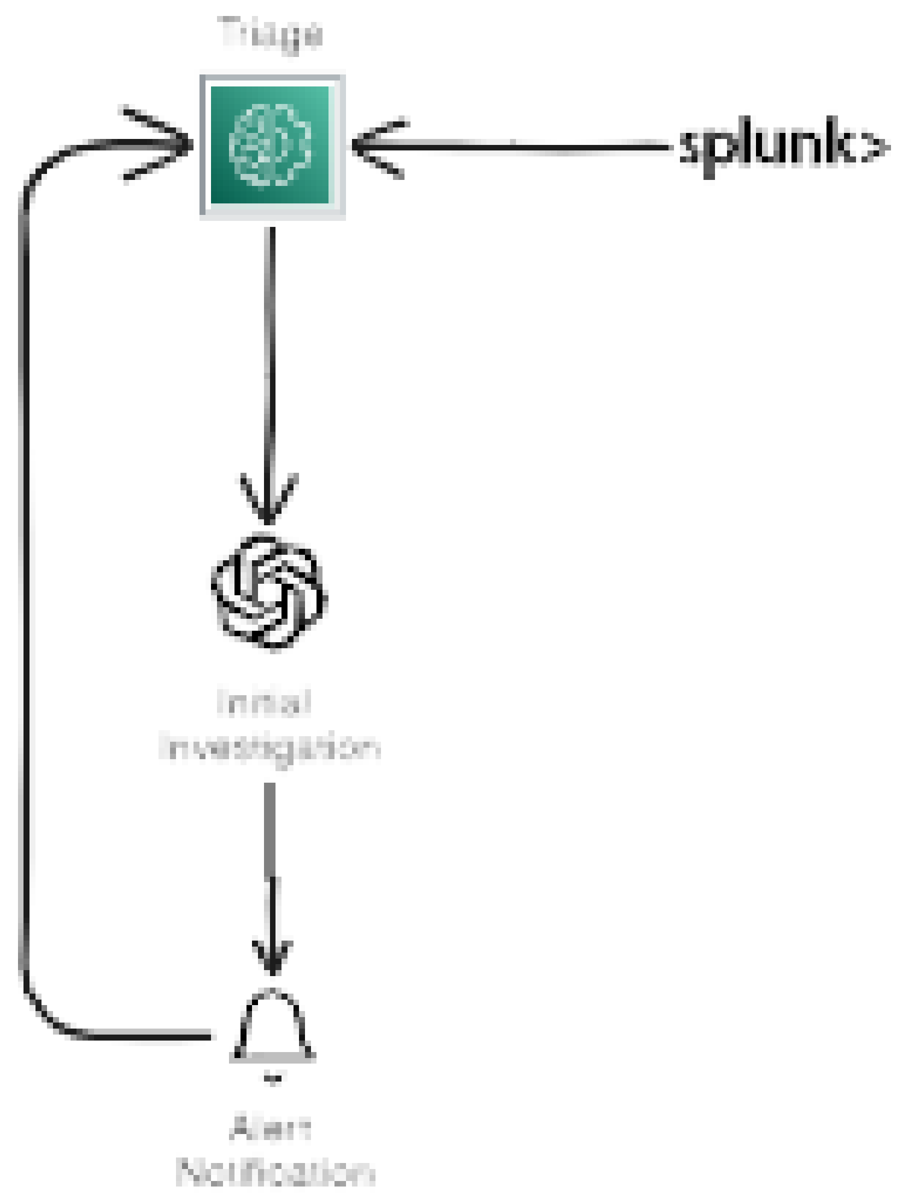
AI in SOC

- **Machine Learning (ML):** ML algorithms analyze vast datasets to improve detection accuracy by adapting to new attack patterns.
- **Natural Language Processing (NLP):** Streamlines the processing of unstructured alerts and incident reports for faster triage.
- **Behavioral Analytics:** Detects deviations from baseline user or device behavior, highlighting insider threats and subtle attacks.
- **Automation & SOAR:** AI automates repetitive tasks—triage, containment, investigation—freeing human analysts for complex, high-priority issues.



How Agentic AI can make it Easy?

- **Detection:** AI correlates security events (Splunk), identifies high-priority incidents
- **Triage:** Automatically prioritizes based on asset criticality and threat severity
- **Analysis:** AI performs initial investigation, gathers evidence, identifies attack vectors
- **Response:** Automated containment executed, stakeholders notified with incident summary
- **Learning:** ML models update detection rules, improve response procedures



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Thank You for Your Time!